

Ex 1 (a) Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.

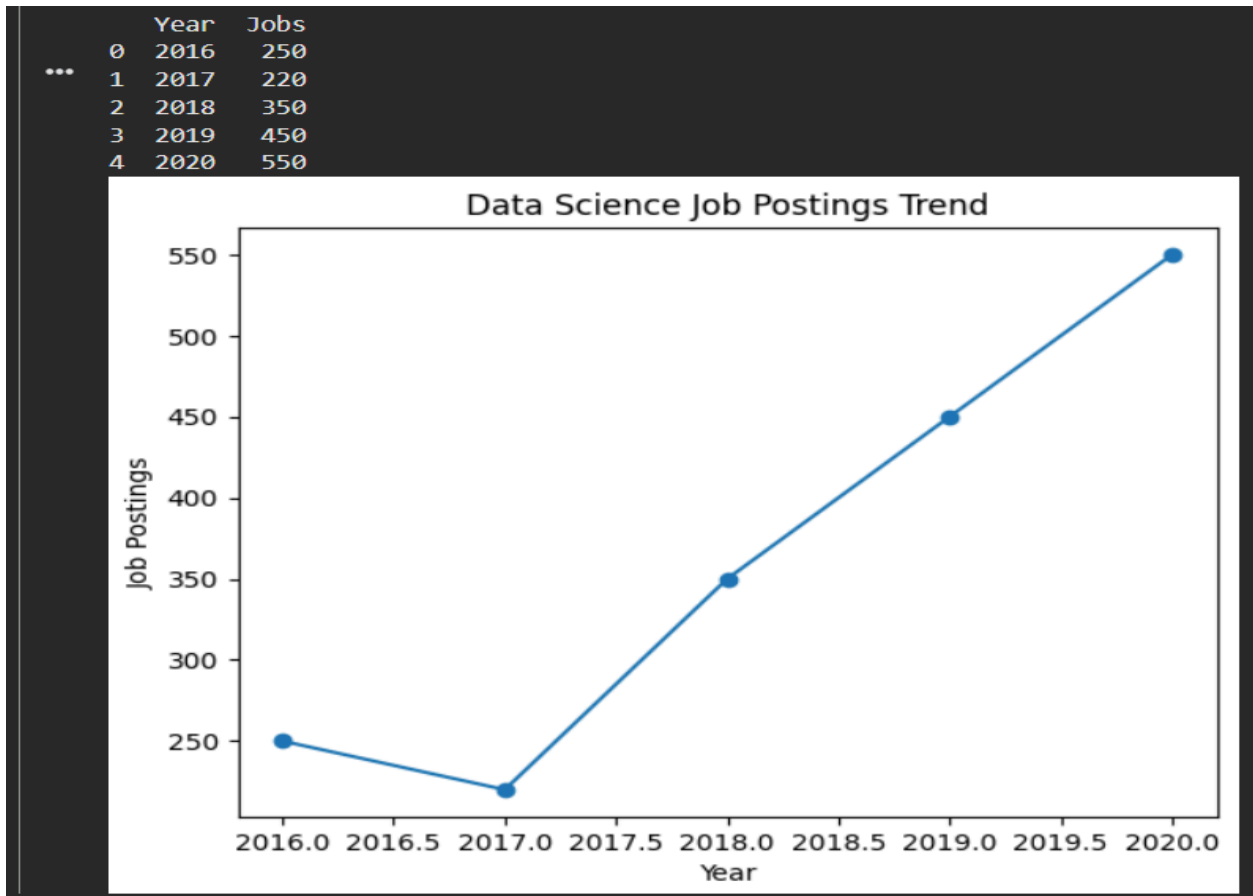
```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Year": [2016, 2017, 2018, 2019, 2020],
    "Jobs": [250, 220, 350, 450, 550,]
}

df = pd.DataFrame(data)

print(df)

plt.plot(df["Year"], df["Jobs"], marker="o")
plt.xlabel("Year")
plt.ylabel("Job Postings")
plt.title("Data Science Job Postings Trend")
plt.show()
```



Ex 1(b) Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Role": [
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Data Analyst"
    ]
}
```

```
df = pd.DataFrame(data)

count = df["Role"].value_counts()

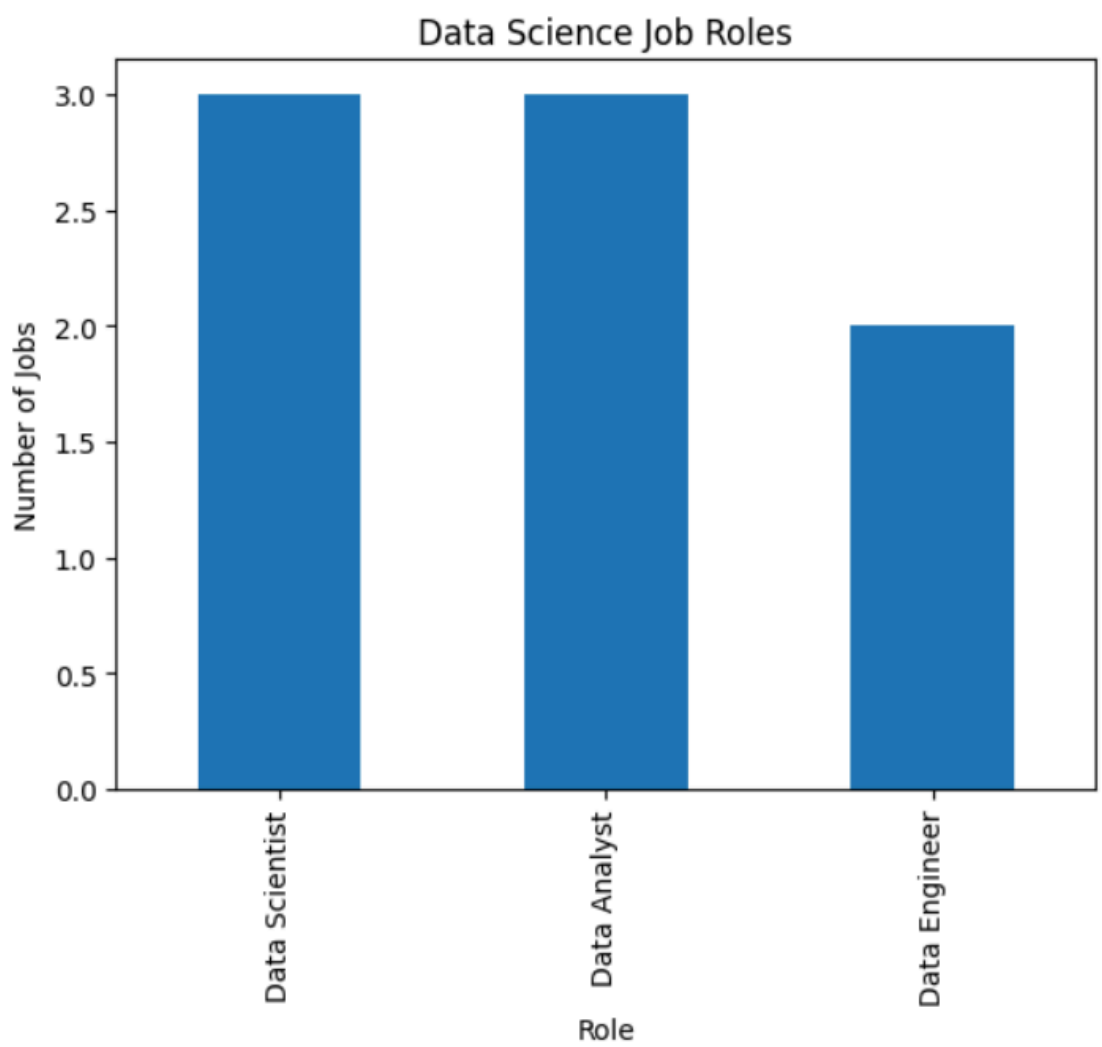
print(count)

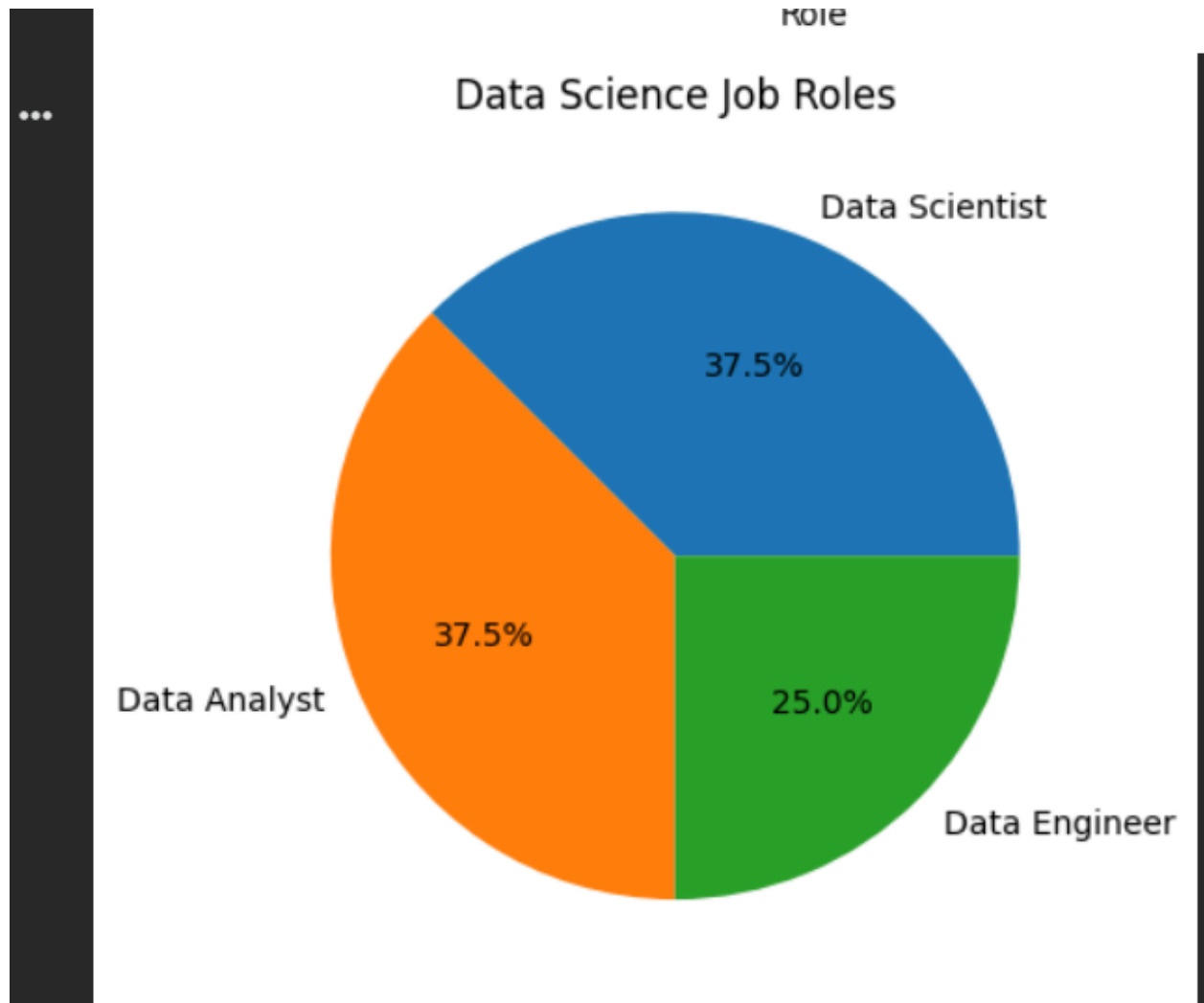
# Bar chart
count.plot(kind="bar")
plt.title("Data Science Job Roles")
plt.xlabel("Role")
plt.ylabel("Number of Jobs")
plt.show()

# Pie chart
count.plot(kind="pie", autopct="%1.1f%%")
plt.title("Data Science Job Roles")
plt.ylabel("")
plt.show()
```

Data Engineer 2
Name: count, dtype: int64

...





Ex 1(c) Conduct an experiment to differentiate Structured, Un-structured and Semi structured data based on data sets given.

```
import pandas as pd

# Structured Data
data = {
    "Name": ["Ram", "Priya", "Arun"],
    "Age": [20, 21, 22],
    "Marks": [80, 90, 85]
}

df = pd.DataFrame(data)

print("STRUCTURED DATA")
```

```
print(df)

# Semi-Structured Data
print("\nSEMI-STRUCTURED DATA")

data = {
    "Name": "Ram",
    "Age": 20,
    "Skills": ["Python", "SQL"]
}

print(data)

# Unstructured Data
print("\nUNSTRUCTURED DATA")

data = [
    "Python is a programming language.",
    "Data Science deals with data.",
    "Machine Learning is part of AI."
]

for x in data:
    print(x)
```

```

STRUCTURED DATA
...
    Name  Age  Marks
0    Ram   20    80
1  Priya   21    90
2   Arun   22    85

SEMI-STRUCTURED DATA
{'Name': 'Ram', 'Age': 20, 'Skills': ['Python', 'SQL']}

UNSTRUCTURED DATA
Python is a programming language.
Data Science deals with data.
Machine Learning is part of AI.

```

Ex 1(d) Conduct an experiment to encrypt and decrypt given sensitive data. Use the cryptography library fernet.

```

from cryptography.fernet import Fernet

# Generate encryption key
key = Fernet.generate_key()
cipher = Fernet(key)

# Sensitive data
data = "Manasa"

# Encryption
encrypted_data = cipher.encrypt(data.encode())

print("Original Data:", data)
print("Encrypted Data:", encrypted_data)

# Decryption
decrypted_data = cipher.decrypt(encrypted_data).decode()

print("Decrypted Data:", decrypted_data)

... Original Data: Manasa
   Encrypted Data: b'gAAAAABqhSeWFOShBRW1W3Cdi6WQLXMyK1yo-MnUuvMsQt8khFYiOShLH
   Decrypted Data: Manasa

```